ASSIGNMENT 10
Design a Model to Detect Fake News from Multimodal Data

1. Aim

To design a basic multimodal fake-news classification pipeline by combining text data and numerical image features using machine learning.

2. Objectives

- To understand fake-news classification.
- To understand multimodal data.
- To convert text into numerical features using TF-IDF.
- To combine text and image-related features.
- To train a classification model.
- To evaluate the model using classification metrics.

3. Theory

Fake-news detection is a classification task that attempts to identify whether a news item is labelled as real or fake according to a dataset's annotation rules.

Multimodal learning combines information from two or more data types, such as text, images, audio, or video.

For example, an online news article may contain a headline, article text, and an accompanying image. Analysing both the text and image may provide more information than analysing the text alone.

This assignment combines two modalities.

1. Text modality: The news headline or article text is converted into numerical features using TF-IDF.

TF-IDF stands for Term Frequency–Inverse Document Frequency. It represents words numerically according to their frequency in a document and their importance across a collection of documents.

2. Image modality: Numerical image features can describe properties such as average brightness and colour-channel values. In a real project, these features should be extracted from the actual image associated with each news article. More advanced systems may use pre-trained image models to obtain image embeddings.

The features from both modalities are combined into one feature matrix. A Logistic Regression classifier is trained on labelled examples to predict the class of a news item.

The dataset in this assignment is artificial and very small. It is intended only to demonstrate feature combination and classification. It cannot establish reliable real-world fake-news detection performance. A practical model requires a larger, labelled dataset containing genuine news text and corresponding images, careful validation, and testing on unseen examples.

4. Algorithm

1. Import the required libraries.
2. Prepare sample news text and class labels.
3. Prepare numerical image features for demonstration.
4. Convert the text into TF-IDF vectors.
5. Combine the text and image features.
6. Divide the dataset into training and testing subsets.
7. Train a Logistic Regression classifier.
8. Predict the test labels.
9. Calculate accuracy and classification metrics.
10. Demonstrate a prediction for a new text-image feature pair.

5.Explanation of Important Code

- TfidfVectorizer() converts text into numerical features.
- fit_transform() learns the vocabulary and converts the sample text into TF-IDF vectors
- csr_matrix() converts the numerical image features into a sparse matrix format.
- hstack() combines text and image features into a single feature matrix.
- train_test_split() divides the dataset into training and testing subsets.
- LogisticRegression() creates the classification model.
- fit() trains the model using the labelled training data.
- predict() predicts whether test examples belong to class 0 or class 1.
- accuracy_score() calculates the proportion of correct predictions.
- classification_report() displays precision, recall, F1-score, and support.

6. Applications

1. Research into text and image combinations in news articles.
2. Educational demonstrations of multimodal machine learning.
3. Analysis of text-image consistency.
4. Media-literacy and misinformation research.
5. Development of larger news-verification systems.

7. Advantages

1. Demonstrates how two different data modalities can be combined.
2. TF-IDF provides a simple baseline for text representation.

3. The pipeline can be extended with real image features.
4. Logistic Regression is relatively easy to implement and interpret.

8. Limitations

1. The artificial dataset is too small for reliable predictions.
2. Simple brightness and colour values do not capture the meaning of an image.
3. A model may learn patterns specific to its training data.
4. Biased or incorrectly labelled training data can produce misleading results.
5. A model's prediction cannot replace proper source checking and fact verification.

9. Conclusion

A basic multimodal fake-news classification pipeline was implemented by combining TF-IDF text features with numerical image features. Logistic Regression was used for classification, and its output was evaluated using accuracy and classification metrics. The experiment demonstrated feature fusion, while highlighting the need for genuine paired data and rigorous testing before developing a real-world system.